Tutorial 4 - Solutions

1. For sample size $n = 40$, we have (by the Central Limit Theorem) $\frac{\bar{X}-\mu}{S/\sqrt{n}} \dot{\sim} N(0,1)$, regardless of the population distribution. So we can use $\frac{\bar{X}-\mu}{S/\sqrt{n}}$ as a pivotal quantity to obtain an approximate 95% CI for μ: $\bar{X} \pm 1.96\frac{S}{\sqrt{n}}$.

   In this example, we have

   $$\bar{x} = \frac{14.152}{40} = 0.3538 \quad \text{and} \quad s^2 = \frac{1}{39}\left(\sum x_i^2 - \frac{(\sum x_i)^2}{40}\right) = 0.0002838.$$

   So, for these data, an approximate 95% CI for μ is given by: $(0.3486, 0.3590)$.

   It is also worthwhile investigating whether the data could have come from a normal distribution. Here is a stem-leaf diagram of the data:

   ```
   The decimal point is 2 digit(s) to the left of the |

   32 | 47
   33 | 00255888
   34 | 116799
   35 | 00222455889
   36 | 2458
   37 | 033578
   38 | 255
   ```

   The normality assumption seems plausible, so could also construct a confidence interval based on the $t-$distribution - i.e., we would assume $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{39}$ and obtain the 95% CI for μ: $\bar{X} \pm t_{39,0.025}\frac{S}{\sqrt{40}} = \bar{X} \pm 2.022\frac{S}{\sqrt{40}}$.

   Assuming normality of the population distribution, we obtain the following 95% CI for μ: $(0.3484, 0.35920)$.

   Note that the CI is wider in this case – this is because we have used the 0.025 percentage point of the t_{39} distribution which has slightly heavier tails than the standard normal.

2. We begin by assuming the normality of the population distribution (we will investigate this assumption below). These data give us:

   $$\bar{x} = 100.8, \qquad s^2 = \frac{1}{19}(214253 - \frac{(2017)^2}{20}) = 570.45, \quad \text{and} \quad s = 23.8841$$

   Under the normality assumption, we obtain the following 95% CI for μ:
   $\bar{X} \pm t_{19,0.025}\frac{S}{\sqrt{20}} = (89.7, 112.0)$ (using $t_{19,0.025} = 2.093$).

   To investigate the normality assumption, we consider the following stem-leaf diagram:

```
 6 | 29
 7 |
 8 | 267
 9 | 446689
10 | 02348
11 | 56
12 | 6
13 |
14 |
15 |
16 |
17 |
18 | 0
```

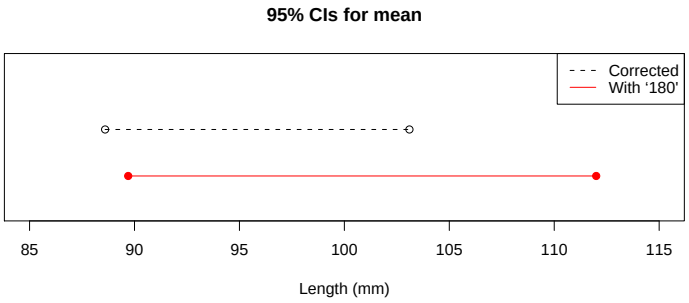
Normality seems plausible, except for the value 180. If 180 is corrected to 80, then the normality assumption is quite reasonable. The confidence interval must be recalculated to reflect this correction. First, we note that

$$\sum x_i = 2017 - 180 + 80 = 1917 \quad \text{and} \quad \sum x_i^2 = 214253 - (180)^2 + (80)^2 = 188253.$$

It follows that the corrected values of $\bar{x}$, s^2, and s are:

$$\bar{x} = 95.85, \quad s^2 = 237.292, \quad \text{and} \quad s = 15.404.$$

The 'corrected' 95% CI for μ is $(88.6, 103.1)$. Diagram for comparison:



3. The main assumption here is that the population is normally distributed. With the observation 180 corrected to 80, this assumption seems reasonable as seen in question 2.

   Using the value of s^2 from the last question we obtain the following 95% CI for σ^2:

$$\left(\frac{19s^2}{\chi^2_{19,0.025}}, \frac{19s^2}{\chi^2_{19,0.975}} \right) = \left(\frac{4508.548}{32.85}, \frac{4508.548}{8.907} \right) = (137.25, 506.18).$$

   So a 95% CI for σ is given by:

$$(\sqrt{137.25}, \sqrt{506.18}) = (11.7, 22.5).$$

4. (a) $\bar{x} = \frac{\sum x_i}{n} = \frac{523}{7} = 74.714$
and
$s^2 = \frac{1}{n-1}\{\sum_i X_i^2 - \frac{1}{n}(\sum_i x_i)^2\} = \frac{1}{6}(39321 - 523^2/7) = 40.905$
Now, assuming that the data come from a *normal population*, a 95% CI for μ is given by

$$\bar{x} \pm t_{6,0.025}\frac{s}{\sqrt{n}} = 74.714 \pm 2.447\sqrt{\frac{40.905}{7}} = 74.714 \pm 5.915$$

i.e. $(68.8, 80.6)$.

(b) If the $86mph$ value is replaced by $76mph$ the sample variance will be *smaller*. With the sample size remaining the same, this will give smaller standard error for the sample mean ($s/\sqrt{n}$), and therefore *narrower* CI for μ.

5. The 95% CI for the population percentage p is $\hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{m}}$, giving

$$|p - \hat{p}| \leq 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{m}}.$$

If we want the margin of error to be less than 0.5% we need to solve

$$0.005 = 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{m}} \Rightarrow m = \frac{1.96^2\hat{p}(1-\hat{p})}{0.005^2}.$$

Now the problem is that we are doing this calculation *before* we gather the data, so we need to have an estimate $\hat{p}$ from some other source. Using the percentage from the previous study, i.e. $\hat{p} = 0.06$ we obtain $m = 8666.6$, so we need a sample of (at least) 8667 people.

Exam Style Question

The data $x_1, \ldots, x_{19}$ below are the scores of 19 randomly chosen candidates in a psychometric test for an accountancy internship.

79	90	101	83	72	76	71
93	91	87	76	86	80	89
77	69	61	86	81		

Note that

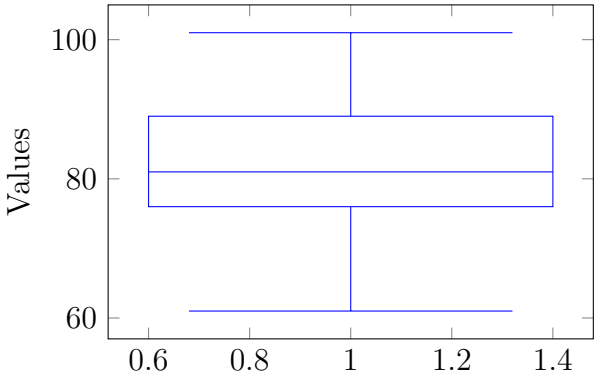
$$\sum_{i=1}^{19} x_i = 1548 \quad \text{and} \quad \sum_{i=1}^{19}(x_i - \bar{x})^2 = 1670.737.$$

(a) By calculating the appropriate numerical statistics, construct a boxplot for the data.

To calculate a box blot we need the following statistics:

$$\begin{aligned} \text{minimum} &: 61 \\ \text{lower quartile} &: 76 \\ \text{median} &: 81 \\ \text{upper quartile} &: \frac{87+89}{2} = 88 \\ \text{maximum} &: 101 \end{aligned}$$

[3 marks]



[1 mark]

(b) In order for $h(\underline{X}, \mu)$ to be a pivotal quantity we need:

- $h(\underline{X}, \mu)$ depends on both the data $\underline{X}$ and the parameter of interest μ;
- The distribution of $h(\underline{X}, \mu)$ to be known or approximately known;
- $h(\underline{X}, \mu)$ is monotonic in the parameter of interest μ.

The first point holds since $h(\underline{X}, \mu)$ depends on μ directly and involves the data via $\bar{X}$ and S.

The second point holds since the distribution of $h(\underline{X}, \mu)$ is approximately t-distributed with $n-1$ degrees of freedom.

The third point holds since $h(\underline{X}, \mu)$ is a linear function of μ and moreover is strictly decreasing in μ.

[3 marks]

(c) We need to assume that $X_1, \ldots, X_n$ is approximately normally distributed, this appears plausible from the box plot.

[1 marks]

Since $h(\underline{X}, \mu)$ is approximately t_{n-1} distributed we have

$$\mathbb{P}(-t_{n-1,\alpha/2} \leq h(\underline{X}, \mu) \leq t_{n-1,\alpha/2}) = 1 - \alpha.$$

Rearranging for μ gives

$$\mathbb{P}\left(\bar{X} - t_{n-1,\alpha/2}\frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} - t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}\right) = 1 - \alpha.$$

Therefore a 95% equal tailed confidence interval is

$$\left(\bar{X} - t_{n-1,0.025}\frac{S}{\sqrt{n}}, \bar{X} - t_{n-1,0.025}\frac{S}{\sqrt{n}}\right).$$

For our data $n = 19$ and

$$\begin{aligned} \bar{x} &= \frac{1548}{19} \\ s^2 &= \frac{1670.737}{18} \\ s &\approx 9.634 \end{aligned}$$

Therefore the observed confidence interval is

$$\left(\frac{1548}{19} - t_{18,0.025}\frac{9.634}{\sqrt{19}}, \bar{X} - t_{18,0.025}\frac{9.634}{\sqrt{n}}\right)$$

From the statistical tables $t_{18,0.025} = 2.101$ giving the observed confidence interval as

$$\left(\frac{1548}{19} - 2.101\frac{9.634}{\sqrt{19}}, \bar{X} - 2.101\frac{9.634}{\sqrt{n}}\right) = (76.830, 86.117).$$

[2 marks]

(d) Since 85 is included in the confidence interval in (c), there is no evidence to reject the assertion that 85 could be the population mean.

[1 marks]

[Total 11 marks]